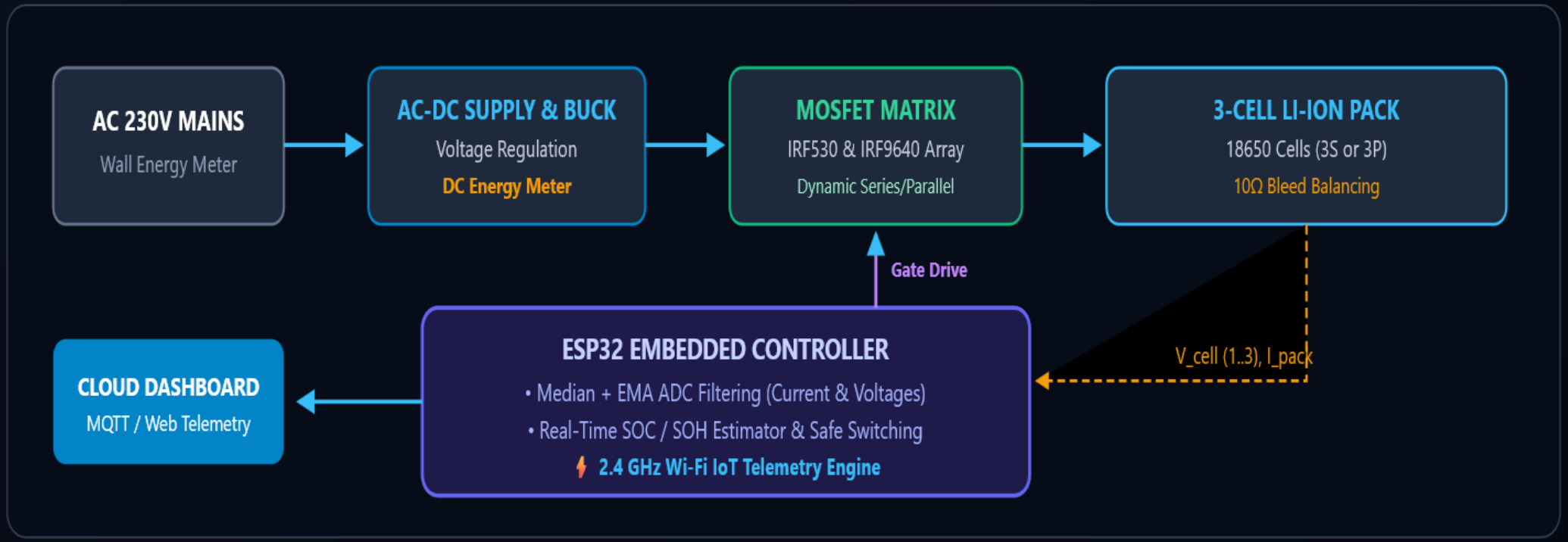
SECTION 3

COMPLETE HARDWARE & SYSTEM ARCHITECTURE

End-to-end hardware pipeline from grid AC power metering to discrete switching and cloud IoT telemetry.



Microcontroller Core

ESP32-WROOM-32 coordinates sensing, filtering, protection rules, and remote telemetry logging over 2.4GHz Wi-Fi.

Dual Measurement Points

Simultaneous AC mains and DC charger power metering isolates upstream adapter losses from pack losses.

Passive Balancing Bleed

10 Ω bleed resistors safely dissipate excess charge from higher-voltage cells to prevent cell divergence.

SECTION 5

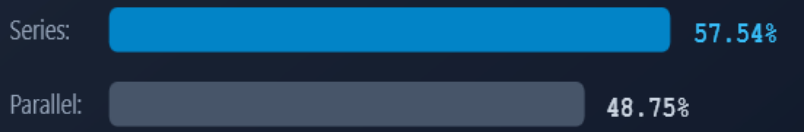
EXPERIMENTAL RESULTS & LOSS ANALYSIS

Empirical validation over two independent charge-discharge cycles with dual-point AC/DC energy normalization.

METRIC	SERIES	PARALLEL	ADVANTAGE
Charge Time (Test 1)	02:18:53	02:24:53	6m Faster
Charge Time (Test 2)	02:10:02	02:28:54	18m 52s Faster
DC Input Energy (Wh)	11.200 Wh	13.220 Wh	15.28% Saved
DC-Side Efficiency	57.54%	48.75%	+18.03% Rel.
AC Wall Energy (Wh)	35.000 Wh	41.967 Wh	16.60% Saved
AC Round-Trip Eff.	18.41%	15.36%	+19.91% Rel.

EFFICIENCY COMPARISON BARS (%)

DC-Side Charging Efficiency



AC Round-Trip Mains Efficiency



Relative Gain in Series: +18.03% (DC) | +19.91% (AC)

1. Conduction Loss Mechanics (I^2R)

For equal power, Parallel draws 3× the cell current.
Because resistive dissipation increases with the square of current ($P = I^2R$),
MOSFET on-state resistance (Rds_on) triggers severe thermal losses in parallel.

Series topology minimizes current to save up to 16.6% energy.

2. Upstream Conversion Bottleneck

Simultaneous DC and AC metering revealed DC efficiency is 49–58%
while AC wall efficiency drops to 15–18%.
This proves the discrete MOSFET reconfiguration stage has low parasitic loss;

the majority of system loss stems from the upstream AC-DC adapter.

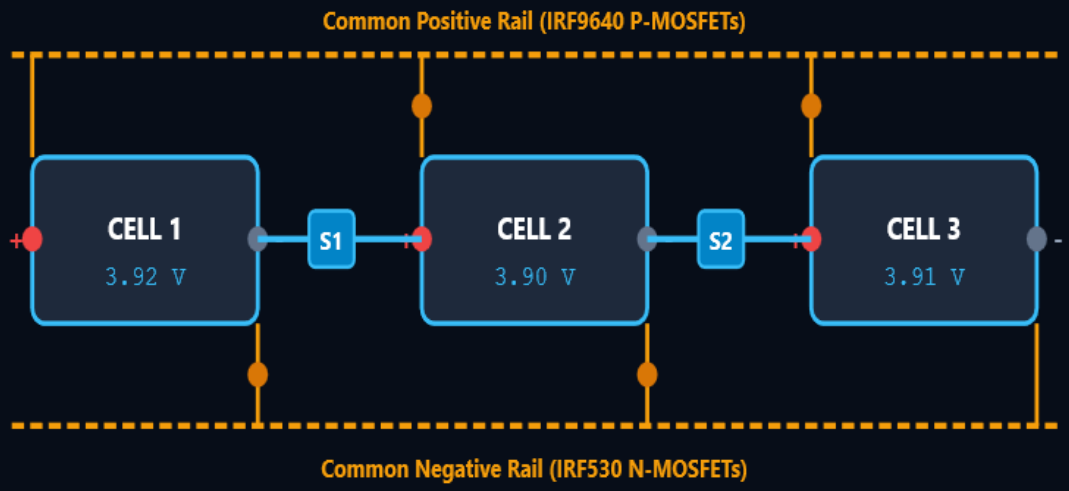
SECTION 2

SWITCHING MATRIX & CIRCUIT SCHEMATIC

Discrete N-channel and P-channel MOSFET network enabling software-driven topology transitions.

DISCRETE RECONFIGURATION MATRIX TOPOLOGY

LTspice Verified



10 Ω Passive Bleed Balancing Network:

Each cell incorporates a 10 Ω shunt bleed resistor switched via firmware during rest to eliminate voltage divergence before parallel mode activation.

Dedicated Gate Driver Level Shifting Stage:

Translates 3.3V ESP32 GPIO logic into full 10V gate drive for reliable IRF530/IRF9640 saturation.

KEY SWITCHING COMPONENTS

IRF530 (N-FET)

Series switches (S1, S2) & low-side rail

IRF9640 (P-FET)

High-side parallel reconfiguration paths

10 Ω Resistors

Per-cell passive balancing bleeder

TOPOLOGY CONTROL TRUTH TABLE

VS = 1, VP = 0	Series Mode (~12.6V, 1x I)
VS = 0, VP = 1	Parallel Mode (~4.2V, 3x I)
VS = 0, VP = 0	Safe Dead-Time (All Open)